

- 1(a) The stored program concept uses the Fetch-Decode-Execute cycle to get the next instruction from memory and then execute it.

Describe what happens during the **fetch** stage of the Fetch-Decode-Execute cycle.

You should state the different registers and buses that are used in your answer.

[4]

- (b) One of the instructions that may be fetched and executed as part of this cycle is a branch instruction.

State the name of the register that would be altered in the **execute** phase during a branch instruction.

[1]

(c) Three ways of improving the performance of a CPU are increasing the clock speed, adding more cores and using pipelining.

Explain how pipelining improves the performance of a CPU.

[3]

(d) The Fibonacci sequence is a series of numbers. It starts with the number 0 and then 1. Each number after that is a sum of the two numbers before it.

The first seven numbers in the series are:

- 0
- 1
- 1 (i.e. 1 + 0)
- 2 (i.e. 1 + 1)
- 3 (i.e. 1 + 2)
- 5 (i.e. 2 + 3)
- 8 (i.e. 3 + 5)

Orla has written some code to show the first five numbers in the Fibonacci sequence (0,1,1,2,3) using the Little Man Computer (LMC) instruction set.

The LMC code that Orla has written contains an error.

START	LDA	MAX
	BRZ	END
	LDA	A
	OUT	
	ADD	B
	STA	B
	LDA	B
	STA	A
	LDA	MAX
	SUB	ONE
	STA	MAX
	BRA	START
END	HLT	
A	DAT	0
B	DAT	1
MAX	DAT	5
ONE	DAT	1

i. State the five outputs that Orla's code would give.

Output 1

Output 2

Output 3

Output 4

ii. Orla has rewritten her LMC code to fix the error and added an additional DAT.

Complete the LMC code to output the first five correct numbers in the Fibonacci sequence (0,1,1,2,3).

START	LDA	MAX
	BRZ	END
	LDA	A
	OUT	
		
	STA	TEMP
		
	ADD	B
	STA	B
		
	STA	A
	LDA	MAX
	SUB	ONE
	STA	MAX
	BRA	START
END	HLT	
A	DAT	0
B	DAT	1
.....	DAT	0
MAX	DAT	5
ONE	DAT	1

[4]

(e) In Orla's LMC code, she used direct memory addressing.

Give **three** other modes of memory addressing.

- 1
- 2
- 3

[3]

(f) Since the development of high level languages, the use of assembly languages has reduced.

Give **two** reasons why in some circumstances programmers will choose to write code in assembly language.

1
.....
2 [2]

(g) The performance of a computer system can be improved by adding more RAM.

Explain why adding more RAM will improve the performance of a computer system.

.....
.....
.....
.....
.....
..... [3]

2(a) An embedded system is often a small device that is designed to carry out a limited number of specialised tasks. Professional athletes sometimes wear small embedded systems called fitness trackers in their shirts. These can be used to track their speed, position, heart rate and other performance data during an event. The tracker transmits this data to a pitch-side server which collates the data from all the athletes.

State the name of **one** input device that might be used in this embedded system and state what it would be used for.

Device
.....
Use
..... [2]

(b) The fitness tracker manufacturers had to decide which type of secondary storage to use.

Explain **two** reasons why flash (solid state) storage would be the best type of secondary storage for the fitness tracker system.

1 .-----

2 .-----

[4]

(c) One use of ROM is to store the Basic Input Output System (BIOS). The BIOS is used when the computer is first turned on.

i. Describe what the BIOS will do to start up the computer.

[3]

ii. Apart from storing the BIOS, ROM can also be used in other ways.

Describe how an embedded system can make a different use of ROM and why it is an advantage.

[2]

i. Describe **two** other roles of an operating system.

2

[4]

Scheduling Algorithm
Round Robin
First come first served
Multi-level feedback queues
Shortest job first
Shortest remaining time

Description
Splits processes into different priority queues based on the amount of processor time they need. It allows them to move between the queues as their characteristics change
Selects the process that takes the shortest amount of time to complete. The processes are run until they are fully complete
Each process is allocated a fixed amount of CPU time. If the process is not complete it will be suspended and the next process will start
Each process is given equal priority and they are processed in the order they arrive
Selects the process that takes the shortest amount of time. The process can be suspended if another shorter process is added

[5]

(e) The fitness trackers will send athletes' performance data to a pitch-side server. A program needs to be developed to analyse this data and display the results.

Different words, colours and charts will be displayed to indicate how well athletes are performing.

The program will be used by athletes from different cultures all over the world.

Discuss the layout considerations that the programmer needs to consider when creating the program user interface for different cultures.

You should include the following in your answer:

- layout considerations
- colour considerations
- character set considerations.

[9]

3(a) State the most appropriate application software for each task in the table.

Task	Application Software
Creating graphics such as a logo	
Writing letters to clients to confirm their appointment date and time	
Calculating the company profits at the end of each month	
Storing, searching and updating client details and purchases	
Creating brochures and flyers about the organisation	

[5]

(b) Application software is run on thin client computers. A thin client computer is a very low-powered computer connected to a powerful central server. The operating system and all the applications run within a virtual machine on the server. The thin client computer will only display the output of the virtual machine and capture and send input to the virtual machine.

State **one** advantage of running the application software within the virtual machine.

[1]

(c) The thin client computer needs to send data that is input by the user to the server so that the virtual machine can process it. For example, it will send text entered by the user on the thin client computer to the application software running on the virtual machine.

Data is compressed in order to improve the speed of data transmission between the client and the server.

Explain why lossless compression should be used instead of lossy compression.

[3]

(d) OCR Solutions make use of software libraries when writing their programming code. They also make use of linkers and loaders to compile and run the program.

i. Describe what **linker** means.

[2]

ii. Describe what **loader** means.

[2]

4(a) Two's complement can be used to represent negative binary numbers.

i. Convert the denary number -124 into an 8-bit two's complement binary number.

[1]

ii. State **one** other way to represent negative binary numbers.

[1]

(b) Convert the denary number 298 into hexadecimal.

[2]

(c) The binary number 10011101 0110 is stored in **normalised** floating point form with an 8-bit mantissa and a 4-bit exponent both written in two's complement.

Convert this binary number into denary.

You must show your working.

[3]

- (d) The binary number 00001011 0111 is represented in **unnormalised** floating point form with an 8-bit mantissa and a 4-bit exponent both written in two's complement.

Convert this binary number into a **normalised** floating point binary number, represented using an 8-bit mantissa and 4-bit exponent format.

You must show your working.

[4]

- (e) Calculate the binary addition of these two 8-bit (unsigned) binary numbers.

Show your working.

$$\begin{array}{r} 00110111 \\ \underline{10011001} \end{array} +$$

[2]

5(a) Ben installs burglar alarms. The alarm is made up of a door sensor and a motion sensor. When the alarm is set, the siren will sound if either the door sensor or the motion sensor detect movement.

The alarm also has a test mode setting. When the test mode setting is enabled, an engineer can check the sensors are working without the siren going off.

The inputs to the alarm are as follows.

Input	System
A	Door sensor
B	Motion sensor
C	Alarm has been set
D	Test mode enabled

Draw a logic circuit to show the logic that is used in the burglar alarm to determine if the siren goes off.

[4]

(b)

- i. Simplify the Boolean expression $\neg A \vee \neg B$ using De Morgan's First Law.

----- [1]

- ii. Simplify the Boolean expression $\neg(\neg B)$ using double negation.

----- [1]

- iii. Simplify the Boolean expression $(A \vee B) \wedge (A \vee C)$ using distribution.

----- [2]

6(a) A company wants more customers to be able to find their website on the internet.

Discuss why search engine indexing and page ranking will be important in achieving this.

You should include the following in your answer:

- what search engine indexing means and what information is collected
- what PageRank means and possible factors that can affect a PageRank score of a website
- possible ways for the company to improve the PageRank score of pages on its website.

[12]

(b) The company would like to start selling their products on their website. They will use both client side and server side processing to do this.

Tick (✓) **one** box on each row to identify whether each task would be best performed on the client side or the server side.

Task	Client Side	Server Side
Loading the website HTML code		
Applying CSS styles to a website		
Running JavaScript code to check that the customer surname has been entered on the order form		
Running queries on the database to check if an item is available in stock		

[4]

(c) When customers contact the company, their computers will use a number of protocols such as TCP/IP and HTTP to make connections with other devices over the internet.

Explain what a protocol is and why they are important in network communications.

[2]

- (d) Describe **three** different pieces of networking hardware in a client-server network apart from the clients and servers themselves.

1

2

[6]

- 7 Rosa has recently opened a new leisure centre. Currently she has 150 members, however, she is estimating that this will increase to around 10 000 members.

Rosa will store members' details in a database. She will also store gym class bookings and appointments with fitness instructors.

Rosa is considering using either a flat file database or a relational database.

Discuss the benefits and drawbacks of flat file databases and relational databases and which one would be the most suitable for Rosa.

You should include the following in your answer:

- the features of flat file databases and relational databases
- the benefits and drawbacks of each type of database
- which type of database would be the most suitable for Rosa.

Blank lined paper for writing.

8(a) Fig. 8 shows a binary search tree that contains the names of different towns in Ireland.

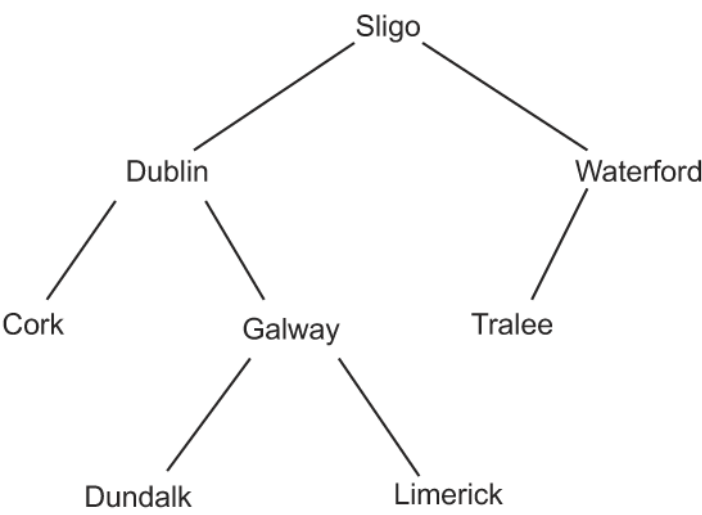


Fig. 8

The binary search tree is held in a 2-dimensional array called towns with 8 rows and 3 columns.

Write a line of program code or pseudocode to declare the array towns.

[2]

(b)

- i. In the 2-dimensional array towns:
- the **first** column contains a pointer to the left side
 - the **second** column contains the data
 - the **third** column contains a pointer to the right side.

Leaf nodes have the pointer null.

Complete the table showing the contents of the towns array to store the binary search tree shown in Fig. 8.

	Left	Data	Right
0		Sligo	
1		Dublin	
2		Cork	
3		Waterford	
4		Galway	
5		Limerick	
6		Tralee	
7		Dundalk	

[4]

ii. Four more towns are added to the binary search tree shown in Fig. 8 in this order:

Mallow

Cavan

Tuam

Wexford

Complete this binary search tree by adding the new towns to it.

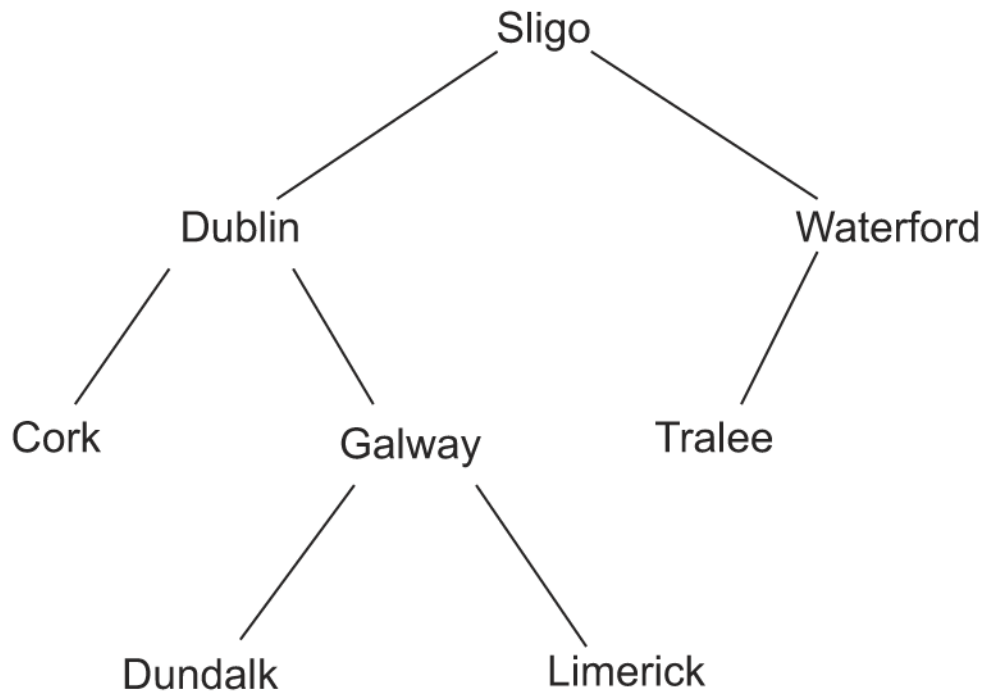


Fig. 8

[4]

- 9 OCR DogCare look after dogs while their owners are at work. They use a program developed in an object-oriented paradigm to store details about each dog.

Each dog is declared by using an instance of the class Dog. This has these private attributes:

- name
- breed
- height
- weight

The constructor method sets all the attributes to the values passed in as parameters.

Write pseudocode or program code to define the class Dog. You should include the attributes and constructor method.

You **do not** need to write the set or get methods.

Blank lined paper for writing.

10(a) Government bodies have been given additional powers under the Regulation of Investigatory Powers Act.

State **three** additional powers that this law gives to some government bodies.

1 .-----

2 .-----

3 .-----

[3]

(b) An investigative firm wants to start investigating cyber security issues. As part of their new roles, employees will be accessing personal data. In order to facilitate this, all employees are enrolled on a course about the Data Protection Act.

Identify **three** principles of the Data Protection Act.

1 .-----

2 .-----

3 .-----

[3]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	1 mark per bullet to max 4: - The contents of the Program Counter/PC are copied/sent to the Memory Address Register/ MAR - The address is sent/transferred along the <u>address bus</u> - The <u>control unit</u> sends/transfers a (read) signal along the <u>control bus</u> - Contents stored in the memory address are sent/transferred along the <u>data bus</u> - Contents (from memory) are stored in the Memory Data Register/MDR ...and sent/copied to the Current Instruction Register/ CIR - The Program Counter/PC is incremented	4	Accept IR/Instruction Register for CIR/Current instruction Register Accept MBR/Memory Buffer Register for MDR/Memory Data Register <u>Examiner's Comments</u> Candidates who had good knowledge of the FDE cycle were able to gain 3 or 4 marks on this question. Some candidates did not have a good understanding of what a register or a bus is and tended to say they did something other than transport or temporarily store. Many however gave clear and correct responses.  Misconception Some candidates gave responses which mentioned a register fetching the data or passing data. A register is a temporary store for data/instructions/addresses and as such doesn't fetch or pass anything. Exemplar 1 <i>The PC holds the next instruction's address which is copied to the MAR, memory address register. The Program Counter is incremented. The address from the MAR is sent on the address bus to the memory. Simultaneously a fetch/read signal is sent on the control bus by the control unit. The data or instruction goes on the data bus and is copied into the Memory Data Register. It is then copied into the CIR.</i> [4] This candidate has shown a clear understanding of the role of the buses and registers in the fetch part of the FDE cycle.
	b	Program Counter // PC	1	<u>Examiner's Comments</u> Candidates who understood the FDE cycle gained the mark. Those that did not tended to mention a register used in the fetch or decode stages.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		1 mark per bullet: - Allows the next <u>instruction</u> to be fetched whilst the previous one is being decoded/executed //allows the overlapping of different parts of the FDE - It increases throughput // increases the number of instructions processed in a set period of time - It prevents the CPU having to wait // prevents idle components	3	DNA responses that talk about the FDE running faster. Allow a diagram demonstrating pipelining for MP1. Allow 'it will take less time to do the same amount of instructions' for MP2 BOD 'processes' for 'instructions' for MP2. DNA 'more efficient' on its own for MP2 DNA points if clearly discussing multiple cores. <u>Examiner's Comments</u> Many candidates were able to gain a mark for an instruction being fetched at the same time as another is decoded. Some candidates discussed multiple cores which were mentioned in the question stem but did not apply to this question. Candidates should have access to previous mark schemes.
	d	i	1 mark per bullet to max 3: 0 1, 2, 4 8 <u>with no numbers after it</u>	3	CAO <u>Examiner's Comments</u> Most candidates were able to gain at least one mark for the first output on this question with many gaining full marks. A small minority of candidates simply copied the outputs given in the question stem.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	1 mark for each correct line. <pre> START LDA MAX BRZ END LDA A OUT <u>LDA B</u> STA TEMP <u>LDA A</u> ADD B STA B <u>LDA TEMP</u> STA A LDA MAX SUB ONE STA MAX BRA STAR T END HLT A DAT 0 B DAT 1 <u>TEMP</u> DAT 0 MAX DAT 5 ONE DAT 1 </pre>	4	CAO Case for mnemonics can be ignored. Case for A, B, TEMP must be all caps. Penalise first error and allow FT. <u>Examiner's Comments</u> Most candidates were able to get the mark for the final answer gap and there was a good range of marks across the cohort.
	e		1 mark per bullet to max 3: • Immediate • Indirect • Indexed	3	DNA direct BOD 'index' for MP3 <u>Examiner's Comments</u> Most candidates were able to give indirect as a mode of memory addressing and there were many candidates who gained full marks on this question. A small number of candidates gave responses like by reference or value, and some gave direct which was mentioned in the question stem.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	f	1 mark per bullet to max 2: • Quicker/more efficient to <u>translate</u> • Makes more efficient use of the CPU // memory // system resources // where a system may have limited resources • The programmer wants direct control over hardware/memory // to access machine specific functionality • Code might be written for a specific architecture • Compilers/interpreters may not be available	2	MP1 must be a comparison e.g. faster/quicker not just fast/quick <u>Examiner's Comments</u> More successful candidates were able to give two valid reasons. Less successful candidates tended to give responses like it is easier to read or easier for the computer to understand. Some candidates were confused between an assembler and compilers/interpreters. The most common correct responses were the programmer wanting direct control over the hardware and it needing to be written for specific hardware.
	g	1 mark per bullet to max 3: • Allows more active/running/temporary data in RAM • It reduces the need to use virtual memory • RAM is faster to access than VM/secondary storage... • ...because data in VM/SS has to be swapped with data in RAM first • Use of RAM rather than VM reduces the risk of disk thrashing • Faster bootup/ shutdown time // reduces load/access time	3	MP1 must imply more/larger amount of data Allow 'instructions/programs/apps/processes/tasks' for data. MP2 - BOD 'no need for Virtual Memory'. <u>Examiner's Comments</u> Most candidates were able to gain 1 or 2 marks on this question. Some confused primary with secondary saying that adding more RAM meant there was less need to use primary memory. Some confused RAM with Cache.
		Total	23	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
2	a	1 mark max for input device and 1 mark max for reason: • Heart rate sensor • Detects athlete's electrical activity / heart rate • GPS (receiver) • Mapping the athlete's movement / position/ speed • Accelerometer/gyroscope/motion sensor • To track the players acceleration/speed/movement/ counting steps/measuring orientation • Button • To allow the athlete to turn it on/off	2	Allow other suitable input devices as long as they are suitable for the scenario and an input device e.g. thermometers, light and UV sensors, skin response sensors, magnetometers, gyrometer, ECG No FT for use if incorrect device Use must be related to the fitness tracker scenario <u>Examiner's Comments</u> The majority of candidates were able to state the name of a device, but many did not apply the use of their named input device to the scenario and gave responses related to a generic use of the device for example giving a heart rate sensor or monitor and then saying the use was to monitor the heart rate with no reference to the fitness tracker. A few gave devices which were unsuitable for an embedded system carried in a shirt such as keyboard and touchscreen.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		1 mark for reason and 1 mark for expansion up to a maximum of 4 marks. • Durable/flash has no moving parts... • ...so will be resistant to damage if the player runs/falls/collides with something • Low power usage... • ...won't need recharging/will last during an event • Small physical size/portable... • ...so it can fit on a shirt/be unobtrusive/be worn // won't affect the athlete's performance • Fast/real-time read/write speed... • ...needed to rapidly record athlete's data	4	Mark across whole answer space Expansion must relate to scenario DNA 'small' on its own DNA anything to do with capacity <u>Examiner's Comments</u> The majority of candidates were able to give a reason such as durable/no moving parts/ small physical size. Candidates who related the reason to why it would be suitable for the fitness tracker were able to gain full marks. Some candidates did not apply their response to the question stem well and instead talked about things like smart watches. Exemplar 2 1. Flash storage has no moving parts so it is more shock resistant than other storage devices (which either require moving components to read or write data). This means that it would be less likely to get damaged during activities like running that the others can get in. 2. Flash storage is also very small in size (but still at the same capacity) so the small device could be smaller. This would make it easier to carry (i.e., it is more portable due to smaller weight and dimensions) and therefore attract/decide to use the others less. [4] The candidate has clearly given two reasons with good application to the scenario.
	c	i	1 mark per bullet to max 3: • Contains the computer start up instructions • Loads settings/configuration (CMOS/NVRAM) • Initialises/checks hardware/peripheral devices are available/work // carry out a POST check... • ...and reports errors • Determines the drive on which the OS is stored • Finds/loads the bootstrap/operating system/kernel (into main memory)	3	DNA 'boots up' on its own <u>Examiner's Comments</u> Many candidates were able to gain at least one or two marks for saying the BIOS finds/loads the operating system and/or checks hardware.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	Mark from one group to max 2: • Storing firmware/ config/ operation instructions/ operating system/ device drivers... • ...that can't be overwritten // doesn't need to be updated • ...so that access is faster OR • Storing security settings/encryption keys/digital signatures... • ...that must be maintained/can't be changed OR • Could be used as <u>primary</u> storage... • ...so no need to load programs into memory • ...so would save time loading up the program OR • Could be used as <u>secondary</u> storage... •so the device will not need additional storage... •to reduce the size of the device	2	<u>Examiner's Comments</u> Strong candidates were able to give a use such as store the operating system and an advantage. Many though, gave a very good description of what ROM is but did not give a different use and so were unable to gain any marks. A few candidates said it could store the BIOS which had already been given in the question. A few candidates talked about microwaves and washing machines which was not relevant to this scenario

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	d	i	2 marks from each group to max 4: • Resource/memory management e.g. • Moving data between RAM and secondary storage/ virtual memory // paging and/or segmentation • Allocating/deallocating memory • Manage hardware/peripherals e.g. • Tracking all devices connected to the system • Device drivers • File management e.g. • Storing files in secondary storage • Searching for //copying // moving // renaming files/folders • Security/user management e.g. • Controlling who can access the system //Managing user profiles • Controlling who can access certain resources on the system // Managing access rights • Provide a user interface e.g. • Allowing the user to interact with the software/hardware/computer • Providing utilities e.g. • Used to monitor // manage // maintain the computer • To manage the security • Providing a platform on which to run software e.g. • Allows additional software to be installed on the computer • To allow the user to complete additional tasks	4	DNA handle interrupts or manage scheduling. Allow any reasonable description that matches with the role. Mark in groups. <u>Examiner's Comments</u> This was generally well answered and the majority of candidates could give two roles. Less successful candidates were then unable to follow through with a reasonable description. More successful candidates were able to give two roles plus relevant and correct descriptions. Some described scheduling and interrupts which were given in the question and gained no marks.

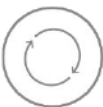
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance												
		ii	1 mark for each correct match. <table><thead><tr><th>Scheduling Algorithm</th><th>Description</th></tr></thead><tbody><tr><td>Round Robin</td><td>Splits processes into different priority queues based on the amount of processor time they need. It allows them to move between the queues as their characteristics change.</td></tr><tr><td>First come first served</td><td>Selects the process that takes the shortest amount of time to complete. The processes are run until they are fully complete.</td></tr><tr><td>Multi-level feedback queues</td><td>Each process is allocated a fixed amount of CPU time. If the process is not complete it will be suspended and the next process will start.</td></tr><tr><td>Shortest job first</td><td>Each process is given equal priority and they are processed in the order they arrive.</td></tr><tr><td>Shortest remaining time</td><td>Selects the process that takes the shortest amount of time. The process can be suspended if another shorter process is added.</td></tr></tbody></table>	Scheduling Algorithm	Description	Round Robin	Splits processes into different priority queues based on the amount of processor time they need. It allows them to move between the queues as their characteristics change.	First come first served	Selects the process that takes the shortest amount of time to complete. The processes are run until they are fully complete.	Multi-level feedback queues	Each process is allocated a fixed amount of CPU time. If the process is not complete it will be suspended and the next process will start.	Shortest job first	Each process is given equal priority and they are processed in the order they arrive.	Shortest remaining time	Selects the process that takes the shortest amount of time. The process can be suspended if another shorter process is added.	5	Examiner's Comments Generally well answered and most candidates gained full marks. Those that did not tended to confuse shortest remaining time with shortest job first but were still able to gain three marks.
Scheduling Algorithm	Description																
Round Robin	Splits processes into different priority queues based on the amount of processor time they need. It allows them to move between the queues as their characteristics change.																
First come first served	Selects the process that takes the shortest amount of time to complete. The processes are run until they are fully complete.																
Multi-level feedback queues	Each process is allocated a fixed amount of CPU time. If the process is not complete it will be suspended and the next process will start.																
Shortest job first	Each process is given equal priority and they are processed in the order they arrive.																
Shortest remaining time	Selects the process that takes the shortest amount of time. The process can be suspended if another shorter process is added.																
	e	Mark Band 3-High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of each of the cultural issues. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. <i>A mark band 3 answer will cover all three points relating to layout, colour and character sets and the cultural impacts of these on the design of the program. The examples will be relevant to the pitch-side program and the needs of the athletes and will go on to evaluate why it is important to ensure these considerations are made.</i> Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of most cultural issues; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. There is a line of reasoning presented with	9	Points may include but aren't limited to: AO1 Knowledge and Understanding Accessibility, colour blindness etc are not relevant to context. Cultural issues are about how different groups of people with particular beliefs, practices or languages may be affected by something. The layout is an important factor to consider. This will determine where items will be placed on the screen. Where items are placed on the screen can massively impact how easily people can use them. The layout which is suitable for one country may not be suitable for another. Colour should be carefully considered. Humans have their own personal perception of different colours and will use this to determine what it means. Colour can be used to confirm messages to the audience, e.g. if our interactions with the software has been successful or not. However, colours can mean different things in different cultures. A character set is a list of characters that can be recognised by the hardware and software. There are various different character sets such as ASCII and UNICODE. AO2 Application Western audiences will often read from left to right and top to bottom. However, other countries around the world read from right													

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	some structure. The information presented is in the most part relevant and supported by some evidence. <i>A mark band 2 answer will cover at least two of layout, colour and character sets and will expand the points, relating these to cultural considerations, the pitch-side program and/or the needs of the athletes although these may not be balanced.</i> Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of some cultural issues; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. <i>A mark band 1 answer will contain some basic, relevant points related to layout, colour and/or character sets. It may not be linked to cultural considerations or the context of the pitch-side program or athlete's needs.</i> 0 mark No attempt to answer the question or response is not worthy of credit.		to left. Athletes are from around the world and this needs to be considered. Traditionally, green often indicates that something is positive or that an interaction with the computer has been successful. Red often indicates something negative or that an interaction with the computer has not been successful. However, these colours may have different meanings and could be seen as offensive in different cultures. Designers therefore need to ensure they don't offend any particular culture as the app is for international athletes. There are many different character sets. The ASCII character set can only represent 128/256 characters which would be suitable for English and European languages. However, this would not be a suitable character set for other world-wide languages and if used, it would mean that characters could not be displayed which would affect the athletes' use of the app. AO3 Evaluation As the software is being used by different cultures, it is important that the user interface can be used and understood by all audiences otherwise people may choose not to use the fitness trackers e.g. using icons rather than words could be more universally understood by the athletes. The programmer needs to consider the typical layout, colours and characters used in different cultures and ensure these are taken into consideration to ensure that groups of people are not offended and will still understand the words, data and charts. Designers could release different versions of the software for different parts of the world or make sure the program has the ability for settings to be changed to overcome any potential issues and allow the athletes to customise their experience. <u>Examiner's Comments</u> There were a good range of Level 2 responses with good discussions for at least two of the bullet points and related to

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					different cultures. Some candidates were unable to apply their knowledge to different cultures and instead discussed accessibility concerns. The candidates with Level 3 responses gave good discussions of all three bullet points and related it to different cultures and the athletes. They were also able to discuss how the programmer could go about this.  Assessment for learning Candidates should be aware that in level of response questions application of knowledge to the scenario is needed to get into the Level 2 and Level 3 mark bands.
			Total	29	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance												
3	a		1 mark for each correct application software <table><tr><th>Task</th><th>Application Software</th></tr><tr><td>Creating graphics such as a logo</td><td>Graphics/image editing software</td></tr><tr><td>Writing letters to clients to confirm their appointment date and time</td><td>Word processing software</td></tr><tr><td>Calculating the company profits at the end of each month</td><td>Spreadsheet/Accounting software</td></tr><tr><td>Storing, searching and updating client details and purchases</td><td>Database software</td></tr><tr><td>Creating brochures and flyers about the organisation</td><td>Desktop publishing software // DTP</td></tr></table>	Task	Application Software	Creating graphics such as a logo	Graphics/image editing software	Writing letters to clients to confirm their appointment date and time	Word processing software	Calculating the company profits at the end of each month	Spreadsheet/Accounting software	Storing, searching and updating client details and purchases	Database software	Creating brochures and flyers about the organisation	Desktop publishing software // DTP	5	Allow any sensible software. e.g. text editor for WP e.g. photo/image editor for row 1 BOD publication editor for DTP DNA brand names - ignore brand name if application type is also given. <u>Examiner's Comments</u> Many candidates were able to score at least 4 marks on this question. Those who gave brand names were not given marks as is standard for these types of question. Few candidates were able to give DTP for the fifth application type. Many gave email for the second application type which would not be the most appropriate for writing letters to clients.
Task	Application Software																
Creating graphics such as a logo	Graphics/image editing software																
Writing letters to clients to confirm their appointment date and time	Word processing software																
Calculating the company profits at the end of each month	Spreadsheet/Accounting software																
Storing, searching and updating client details and purchases	Database software																
Creating brochures and flyers about the organisation	Desktop publishing software // DTP																
	b		1 mark for: • Reduced hardware is needed on computers // reduced hardware costs • Improved security by keeping all software running from one physical device • Easier/cheaper to manage as only one physical device runs the programs. • Can add/remove resources/memory/processes to the VM • If it is infected by malware it can be deleted and recreated // the rest of the system isn't affected/is protected against malware • No direct access to hardware • Resources can be used more flexibly between the machines	1	<u>Examiner's Comments</u> There were some good responses to this question with many giving the response that Malware would not affect the server. Some talked about being able to try out different operating systems which was not relevant to this scenario.												

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		1 mark per bullet to max 3: • Lossless will not permanently remove data • Lossless can be fully reconstructed/restored • Quality (of text/graphics/sound) is not lost • Any loss of text would be noticeable/would make it unreadable/unusable • Lossless rewrites data in a more efficient format	3	Accept reverse points e.g. lossy will delete data permanently <u>Examiner's Comments</u> This is a common topic in previous papers and most candidates were able to gain at least one mark with more successful candidates gaining all three. Some candidates gave responses that were too vague and did not relate to the question stem.
	d	i	1 mark per bullet to max 2: • Combines/links code/programs to files/software libraries... • ...to form a single executable file	2	Allow one mark for valid description of static and/or dynamic linkers e.g. Static linkers combine code and libraries into one file // Dynamic linkers link/add addresses to libraries <u>Examiner's Comments</u> The majority of candidates were able to gain one mark on this question for stating that a linker links code with libraries but few went on to then describe it forming a single executable file.
		ii	1 mark per bullet to max 2: • It is part of the operating system • Loads an executable file (into memory)... • ...from secondary storage • Loads the required software libraries	2	MP2 - Allow loads applications/programs <u>Examiner's Comments</u> Many candidates confused loaders with compilers and IDEs and talked about being able to use libraries in source code or loading code into the compiler.
			Total	13	

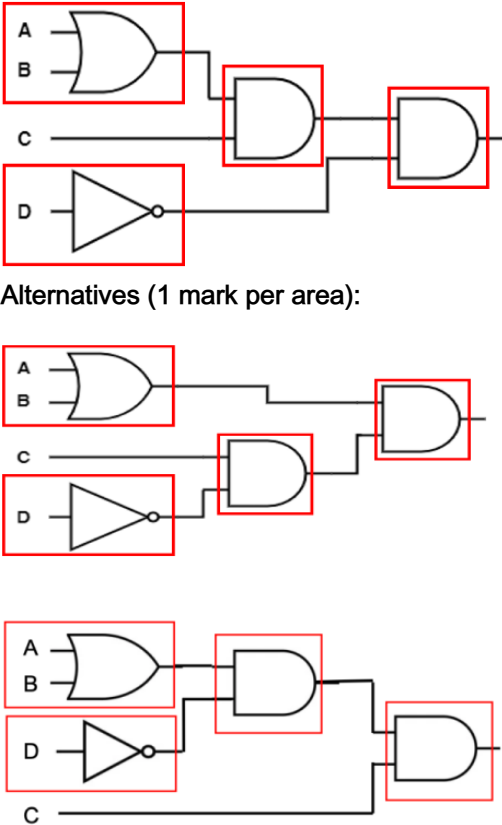
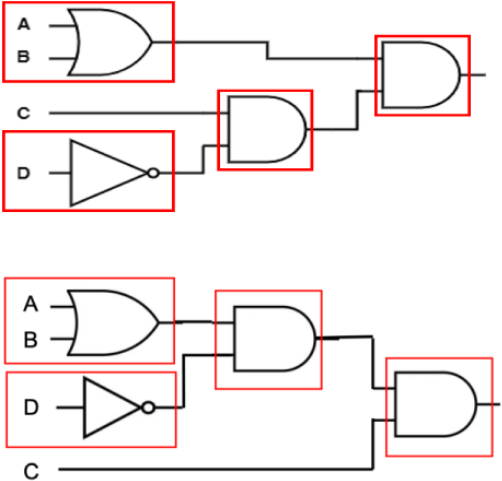
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a	i	1000 0100	1	CAO <u>Examiner's Comments</u> This was generally well answered with the majority of candidates gaining full marks. Those that did not had either got it completely wrong or had missed or added an extra zero.
		ii	Sign and magnitude	1	Accept: One's complement <u>Examiner's Comments</u> This was generally well answered and the majority of candidates gained one mark.
	b		12A • 1 mark for 12 • 1 mark for A	2	Must be in correct order. <u>Examiner's Comments</u> This was generally well answered and the majority of candidates gained two marks.
	c		1 mark per bullet to max 3: • Converting exponent to 6 • Converting mantissa to 1001110.1 // moving binary point 6 to the right • -49.5	3	MP2 - if flipped = 110001.1 Correct answer, with any binary working gets full marks <u>Examiner's Comments</u> The majority of candidates were able to gain a mark for converting the exponent to 6 and many were able to gain the second mark for moving the point 6 places to the right. Many candidates gained full marks although some were unable to convert the binary number back to denary to give -49.5
	d		1 mark per bullet to max 4: • Reducing exponent to 4 // reducing exponent by 3 • New mantissa starts 01011... • ... and is written out to 8 bits making 01011000 • New exponent of 0100	4	Correct mantissa and exponent, with any binary working gets full marks <u>Examiner's Comments</u> There were a range of responses with more successful candidates gaining full marks. Some candidates lost marks for incorrectly writing the final mantissa to 8 bits.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	e		1 mark per bullet to max 2: • 1101 0000 • Correct working out/ 6 carries shown	2	0 mark for denary calculations with no carries shown. <u>Examiner's Comments</u> The majority of candidates were able to gain full marks on this question showing accurate working out and a correct response.
			Total	13	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
5 a	1 mark for each correct area including labelled inputs  Alternatives (1 mark per area): 	4	Max 3 if any additional gates. NOT gate must have only one input, AND/OR gates must have two inputs NOT gate must have circle Ignore any names on gates Examiner's Comments There were a range of marks on this question. More successful candidates gave all four gates accurately, but most candidates were able to gain at least one or two marks. Some candidates were unaware of what a logic circuit was and instead produced a range of different diagrams including flow charts.
b	i	1	Accept NOT (A AND B) / other correct notations e.g. $\overline{A \cdot B}$ $\neg$ must be the correct way round Examiner's Comments Many candidates gained full marks with the most common error being $\neg(A \vee B)$
	ii	1	Examiner's Comments Almost all candidates were able to gain a mark on this question.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	• 1 mark for A • 1 mark for $\vee (B \wedge C)$ $A \vee (B \wedge C)$	2	Doesn't need brackets Allow other correct notation e.g. $A+B.C$ Allow either order e.g. $(B \wedge C) \vee A$ Allow $A \vee (BC)$ / $A \vee BC$ <u>Examiner's Comments</u> The majority of candidates recognised that distribution meant that there was only one A following simplification but the most common error for the second mark was again to use OR instead of AND for B AND C
			Total	8	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
6	a	Mark Band 3-High Level (9-12 marks) The candidate demonstrates a thorough knowledge and understanding of search indexing and page rank. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate is able to assess the extent to which page rank and search engine optimization is important to online visibility. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. <i>A mark band 3 answer will explain in detail how search indexing happens and the factors affecting PageRank scores. There will be clear evaluative points that identify how to improve PageRank score.</i> Mark Band 2-Mid Level (5-8 marks) The candidate demonstrates reasonable knowledge and understanding of search indexing and page rank; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate makes a reasonable attempt to come to a conclusion as to why page rank and search engine optimisation are important to online visibility. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. <i>A mark band 2 answer will cover both search indexing and page ranking and will expand the points, explaining what</i>	12	Points may include but aren't limited to: AO1 Knowledge and Understanding Search engine indexing: • Search engine indexing is where a database of key words is kept with links to relevant pages stored with the keyword • Words are stored along with their position on a page • Bots (spider/crawlers) will "crawl" the web finding web pages and looking for key words on them • They will navigate from page to page following hyperlinks • They are then sent back to the search engine's database • When you search for a website, you are not searching on the web, you are searching in the search providers database PageRank: • Used to find and rank website pages and then list the results in a search engine. If a website has a higher score, it will appear higher in the list of search results. • More links from more important pages, ranks a page higher • Stored as a weighted, directed graph ◦ Pages are nodes ◦ Hyperlinks are edges in one direction ◦ Weightings are calculated by page rank AO2 Application To get more of presence they will need to... • Make effective use of meta tags that effectively describe the contents of the website. • Make effective use of H1 tags using suitable headings that describes the content of the website • Increase the number of quality of incoming links from other websites, ideally those with a high PageRank score themselves • Increase the number of outgoing links • Key words can be stored in meta tags,

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	<i>information is collected during indexing and factors that affect PageRank scores although these may not be balanced. There should be some attempt to identify how to improve PageRank score.</i> Mark Band 1-Low Level (1-4 marks) The candidate demonstrates a basic knowledge of search indexing or page rank; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. <i>A mark band 1 answer will contain some basic, relevant facts about what search indexing is and/or what PageRank means. They may not discuss how the company can improve PageRank.</i> 0 mark No attempt to answer the question or response is not worthy of credit.		title tags or header tags (h1, h2 etc) • Damping factor is used to quash a PageRank based on the idea that a user will only click through a certain number of links AO3 Evaluation • PageRank algorithm is only one algorithm that is used to Rank websites. • To get a better presence on the internet, the company will need to consider who links to their page, the more popular the better • They could potentially work with other companies who have established websites and get incoming links from those sites • It's important that a company ensures their site is malware free/secure as this can impact on their PageRank score <u>Examiner's Comments</u> Many candidates were able to discuss some factors that affect PageRank and how the website could improve it. Far less were able to discuss search engine indexing correctly. Some were able to apply their knowledge to the scenario. More successful candidates were able to give a good description of both search engine indexing and PageRank as well as relevant suggestions for how the website could improve their PageRank.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance															
	b	1 mark for each correct row. <table><thead><tr><th>Task</th><th>Client Side</th><th>Server Side</th></tr></thead><tbody><tr><td>Loading the website HTML code</td><td>✓</td><td></td></tr><tr><td>Applying CSS styles to a website</td><td>✓</td><td></td></tr><tr><td>Running JavaScript code to check that the customer surname has been entered on the order form</td><td>✓</td><td></td></tr><tr><td>Running queries on the database to check if an item is available in stock</td><td></td><td>✓</td></tr></tbody></table>	Task	Client Side	Server Side	Loading the website HTML code	✓		Applying CSS styles to a website	✓		Running JavaScript code to check that the customer surname has been entered on the order form	✓		Running queries on the database to check if an item is available in stock		✓	4	Accept alternatives to ticks in the boxes so long as it is clear which side is marked. <u>Examiner's Comments</u> Most candidates were able to gain 2 or 3 marks on this question with many gaining all 4 marks. The most common error was ticking server side for Running JavaScript to check the surname.
Task	Client Side	Server Side																	
Loading the website HTML code	✓																		
Applying CSS styles to a website	✓																		
Running JavaScript code to check that the customer surname has been entered on the order form	✓																		
Running queries on the database to check if an item is available in stock		✓																	
	c	1 mark per bullet to max 2: 1 mark for: • A set of <u>rules</u> for communication/ transmission/transfer of data between devices 1 mark max from: • The sending system has no direct control over the way the receiving system responds • Allows standardisation as different devices have different OS/hardware • Protocols allow for data to be exchanged in a predictable way/ in the correct format	2	Not 'a rule' - must be plural MP1 not a set of instructions MP1 accept standards / an agreement <u>Examiner's Comments</u> Most candidates were able to explain that a protocol is a set of rules for communication between devices. Few then went on to explain what that meant and so the majority of candidates gained one mark.															

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	d	1 mark for hardware and 1 mark for expansion to 6 marks max • Router e.g. • Connect networks together • Assign IP address to devices • Examines data packets and forwards them • Cable/ Ethernet e.g. • Carries digital data from one device/NIC to the next • Connects wired devices to the network • Gateway e.g. • Connects different types of network • Translates protocols from one network to another • Bridge e.g. • Connects different network segments • Repeater • Receives a signal and retransmits it • Network Interface Card // NIC e.g. • Gives each device a MAC address / unique ID • Allows a computer system to interface with a network • Wireless Access Point // WAP e.g. • Allows wireless devices to communicate with each other • Sends and receives radio waves • Examines data packets and forwards them • Switch e.g. • Connects multiple wired devices to the network • Receives data and forwards it to the intended recipient • Examines data packets and forwards them • Routes based on MAC addresses • Hub e.g. • Receives data from a device and broadcasts it to all devices connected to it • Modem e.g. • Changes a signal from digital to analogue	6	2 marks max each. Allow any suitable expansion. Mark in pairs. Allow: Proxy server e.g. • Sits between user and computer to route requests through an external server • Creates separation between a user and the site/service • Protects your security/anonymity by hiding IPs <u>Examiner's Comments</u> The majority of candidates could give three different pieces of networking hardware but many were unable to then describe what they did with many putting connects devices for the description of each piece of hardware.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			• Firewall e.g. • Filters traffic coming in and out of a network		
			Total	24	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
7	Mark Band 3-High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of databases. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. <i>A mark band 3 answer will cover benefits and disadvantages of both flat file and relational databases. The expansions will relate to Rosa's needs and will go on to evaluate why a relational database is most appropriate.</i> Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of databases; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. <i>A mark band 2 answer will cover features of both flat file and relational databases and will expand the points, relating these to benefits and drawbacks and size of Rosa's data set although these may not be balanced. There will be an attempt to recommend which type is suitable.</i> Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of databases; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to	9	Points may include but aren't limited to: AO1 Knowledge and Understanding A flat file database stores data in a single table, often in a plain text file. Usually, each line will hold a single record and delimiters such as commas are used to separate this into different fields. Benefits: • They are easy to set up as only one table is needed • All records are stored in one place Drawbacks: • Tables can contain lots of redundant data which increases the amount of storage space needed • Searching the database can be slower as there is more data to search • The database can be more difficult to manage and possibly expand in the future A relational database stores data in multiple different tables. These are linked together using relationships created by primary and foreign keys. Benefits: • There is less duplication of information so less redundant data which can reduce the file size • The database is easier to manage and easier to expand in the future • Searching the database may be faster as there is less data / not all tables need to be searched • It can be more secure as different tables means different people could have different access Drawbacks: • They are generally more difficult to set up and need more technical knowledge to do things like normalisation. • Data is spread out between multiple tables. Some tables may be link tables which only display key fields /

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. <i>A mark band 1 answer will contain some basic, relevant points related to features of flat file and/or relational databases. It may not be linked to Rosa's specific needs.</i> 0 mark No attempt to answer the question or response is not worthy of credit.		references • They are more expensive as you will need a DBMS rather than just a basic file AO2 Application • Given that Rosa currently has 150 members, a flat file database will be suitable • However, as the business expands, using a flat file database may prove problematic • She could end up with a lot of redundant data as each time a new booking/appointment is made, it may store repeated data. • The more data Rosa has, the more likely there could be inconsistencies if data is edited/updated and her data would lose integrity • A relational database would be more efficient. • Membership details can be stored once and then their membership ID number can then be used to make a booking • A relational database will also allow her to expand even further in the future as additional tables can be added. However, the database will need to be normalised to ensure data is consistent, in the same format and that dependencies are reduced AO3 Evaluation The most suitable database for Rosa would be a relational database. Although this is more complex and the data needs to be normalised, it will store less repeated data when her company expands. It will also allow the database to be easily maintained and expanded in the future and will be easier to keep data secure, which is important. <u>Examiner's Comments</u> There were some really good responses to this question. More successful candidates were able to explain the features of both types of database as well as giving relevant benefits and drawbacks of each which they then applied to the scenario well and gave a recommendation. Some candidates gave good descriptions of the

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			features of each type but were unable to apply their knowledge to the scenario other than saying the membership was going to increase. There were a number of candidates who confused a database with a table stating that a relational database was a number of related databases and not a number of related tables.  Misconception The main misconception was to confuse database with table although some candidates also confused the different keys used in a relational database and what they were used for. Exemplar 3 A flat file database is a type of database that only contains one table. This means that all data is stored in a table, made up of records and fields. It doesn't require a primary key, or it can be normalised. A relational database uses multiple tables linked together using foreign keys, which creates one to many relationships between the different tables in the database. A flat file database has many advantages. Firstly, it is simple to understand, add, create and modify. This is because all records are stored in one table, so all the information is in one place. However, there are drawbacks to this. If a large number were to book multiple classes and appointments, extra records would be created. This would result in a large amount of redundant data being included, such as a member's details appearing more than once. This not only increases the amount of space the database takes up, but also makes it more difficult to search the database. A relational database also has many benefits. When a relational database has been normalised, it becomes very efficient to search and locate records within it. As well as this, through the process of normalisation, large amounts of redundant data would be deleted, reducing the storage space required for the database. For example, by creating tables for appointments and another for classes, and another for member details, the membership details wouldn't need to be written out again each time an appointment is booked, saving Rosa time. Overall, a relational database would be much more suitable for Rosa. Although a flat file database is fine for Rosa to use with 150 members, if she has 10,000 members in her leisure centre it will be incredibly difficult to locate specific records when searching, as there would be far too much redundant data, therefore and the file size would be incredibly large if extra details are included each time an appointment is made. However, if Rosa were to use a relational database, there would be much less redundant data, therefore reducing the file size and making it easier for her to find records. Additionally, secondary keys and indexing will be used on a relational database to speed up the search process, although this may not be possible with 10,000 members due to the file size. Therefore, she should use a relational database. The candidate has shown good application to the scenario as well as benefits and

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					drawbacks of both database types. They have made a valid recommendation which is justified.
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																																									
8	a		• Typed as array/string and set to the variable towns • 2 dimensions set to 8 and 3 in brackets Examples array towns(8,3) string towns[][] = new string [8][3] array towns [8][3] as string	2	For MP1 accept any suitable attempt at typing as an array, for example towns = [], towns() For MP2 accept any suitable attempt to set to 2 dimensions of the correct size Allow 7, 2 as array index Allow column major notation i.e. array towns [3,8] <u>Examiner's Comments</u> Many candidates were able to gain full marks with most being able to gain one. Many candidates did not specify the number of elements in the array which was given in the question and tended to give responses such as towns = [] and as such only gained one mark.																																									
	b	i	<table><thead><tr><th></th><th>Left</th><th>Data</th><th>Right</th><th>Marking Guidance</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>Sligo</td><td>3</td><td rowspan="2">1 Mark</td></tr><tr><td>1</td><td>2</td><td>Dublin</td><td>4</td></tr><tr><td>2</td><td>null</td><td>Cork</td><td>null</td><td rowspan="2">1 Mark</td></tr><tr><td>3</td><td>6</td><td>Waterford</td><td>null</td></tr><tr><td>4</td><td>7</td><td>Galway</td><td>5</td><td rowspan="2">1 Mark</td></tr><tr><td>5</td><td>null</td><td>Limerick</td><td>null</td></tr><tr><td>6</td><td>null</td><td>Tralee</td><td>null</td><td rowspan="2">1 Mark</td></tr><tr><td>7</td><td>null</td><td>Dundalk</td><td>null</td></tr></tbody></table>		Left	Data	Right	Marking Guidance	0	1	Sligo	3	1 Mark	1	2	Dublin	4	2	null	Cork	null	1 Mark	3	6	Waterford	null	4	7	Galway	5	1 Mark	5	null	Limerick	null	6	null	Tralee	null	1 Mark	7	null	Dundalk	null	4	Allow -1 or Ø for null FT for 0/blank used as null after first error. <u>Examiner's Comments</u> The question states that the first and third column contain a pointer and the second contains the data. The biggest error by far was candidates putting data in the first and third column instead of a pointer which meant that many only gained a mark for having null in the last two rows. More successful candidates were able to gain full marks.
	Left	Data	Right	Marking Guidance																																										
0	1	Sligo	3	1 Mark																																										
1	2	Dublin	4																																											
2	null	Cork	null	1 Mark																																										
3	6	Waterford	null																																											
4	7	Galway	5	1 Mark																																										
5	null	Limerick	null																																											
6	null	Tralee	null	1 Mark																																										
7	null	Dundalk	null																																											
		ii	1 mark for each correct town. <pre>graph TD Sligo --> Dublin Sligo --> Waterford Dublin --> Cork Dublin --> Galway Cork --> Cavan Galway --> Dundalk Galway --> Limerick Waterford --> Tralee Waterford --> Wexford Tralee --> Tuam Limerick --> Mallow</pre>	4	If same town added more than once BOD if one in the right place Ignore basic spelling mistakes and case Towns must be to left or right of the node above - DNA nodes straight below. <u>Examiner's Comments</u> This was generally well answered and many candidates gained full marks with many candidates writing out the alphabet beneath the question to aid then in placing the new towns.																																									
			Total	10																																										

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
9	1 mark per bullet to max 5: • Suitable logic for class dog declaration • Suitable logic to define the 4 (private) attributes: name, breed, height, weight • Suitable logic to declare a <u>public</u> method for constructor (e.g. new or class name)... •taking <u>only</u> 4 different parameters in any order. • Suitable logic to set the values of each attribute MP4 - if it is obvious the candidate has used Pythonic syntax: allow the additional 5th parameter representing the object e.g. self, turtle <u>but this must be the first parameter listed.</u> MP2 - If they have done the above BP2 can be considered implicitly met (Python doesn't require attributes to be declared outside the constructor)	5	Mark in a vertical line against each MP. Ignore data types in attribute names Allow colon/empty brackets at the end of class def Example Solutions <u>Pseudocode style</u> <pre>class Dog private name private breed private height private weight public procedure new (nameIn, breedIn, heightIn, weightIn) name = nameIn breed = breedIn height = heightIn weight = weightIn endclass</pre> <u>Java / C# Style</u> <pre>Class Dog { private String name private String breed private float height private float weight public void Dog(String name, String breed, float height, float weight) { this.name = name this.breed = breed this.height = height this.weight = weight } }</pre> <u>Python Style</u> <pre>class Dog: def __init__(self, name, breed, height, weight): self.name = name self.breed = breed self.height = height self.weight = weight</pre> <u>Examiner's Comments</u>

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					There were a range of responses to this question with the most common responses being written in pseudocode or python. There was a clear division between candidates who were comfortable writing Object Oriented Programming (OOP) code and class constructors and those that were not. Less successful candidates with little OOP experience often confused the class declaration with the constructor declaration. A few input the values rather than setting the attributes to the values passed in the parameters.
			Total	5	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
10	a	1 mark per bullet to max 3: • Allows for mass surveillance of communications • Allows the monitoring of an individual's internet activities/history • Allows covert surveillance to be carried out • Can demand access to protected data • Can demand that ISPs/businesses give access to customer communications/history • Can demand that ISPs/businesses install equipment that facilitate surveillance • Can demand that encryption keys are handed over // force individuals to decrypt data • Can keep existence of searches and what found private in court	3	Allow other suitable answers relating to examples of other types of surveillance e.g. CCTV, directed surveillance. MP4 relates to password protected data or accessing data protected under the DPA <u>Examiner's Comments</u> There were a number of candidates who could not accurately name one additional power.
	b	1 mark per bullet to max 3: • Data must be processed fairly/ lawfully • Data must be adequate/ relevant/ not excessive • Data must be accurate and (where necessary) up to date • Data must not be retained for longer than necessary • Data can only be used for the purpose for which it was collected • Data must be kept secure • Data must be processed in accordance with people's rights • Data must not be transferred outside of the EU without adequate protection // to other countries who do not have equivalent data protection laws	3	Accept: Right to see their own data/change if incorrect <u>Examiner's Comments</u> This was generally well answered and candidates showed good understanding of the Data Protection Act (DPA) principles.
		Total	6	